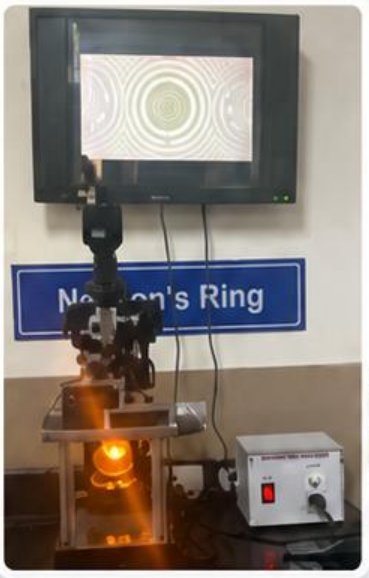
MEDICAPS UNIVERSITY

















APPLIED PHYSICS LAB MANUAL

Faculty of Science
Department of Physics

DIFFRACTION GRATING

To determine the wavelength of spectral lines of mercury vapour lamp with the help of Grating and Spectrometer

5.1. Objective: To determine the wavelength of spectral lines of mercury vapour lamp with the help of Grating and Spectrometer

5.2. Apparatus/Components Required: Spectrometer, grating, mercury lamp, power supply, magnifying glass, and table lamp.

5.3. Theory: Diffraction and Fraunhofer Diffraction: The spreading-out (bending or deviation from its original path) of a wave when it passes through a narrow opening is usually referred to as diffraction and the intensity distribution on the screen known as the diffraction pattern.

In the Fraunhofer diffraction, the source and the screen are at infinite distance from the aperture or slit, this is easily achievable by placing the source on the focal plane of a convex lens and placing the screen on the focal plane of another convex lens

The two lenses effectively moved the source and screen to infinity because the first lens makes the light beam parallel and second lens effectively makes the screen receives a parallel beam of light. Further, the Fraunhofer diffraction pattern is not difficult to observe; all that one needs is laboratory spectrometer. The collimator renders a parallel beam of light, and the telescope receives parallel beam of light on its focal plane. When the value of N is very large, one obtains intense maxima, when

$$dsin\theta = n\lambda \quad \textbf{(n=0, 1, 2)}$$

Components of Spectrometer:

Diffraction Grating:

An arrangement which essentially consists of many equidistant slits is known as a diffraction grating; the corresponding diffraction pattern is known as the grating spectrum. The principal maxima corresponding to different spectral lines will

correspond to different angles of diffraction. Thus, the grating spectrum provides us with an easily obtainable experimental set up for determination of wavelengths.

Spectrometer:

The spectrometer is an instrument which is usually employed for the study of the composition of light and the measurement of refractive indices of solids and liquids. It consists essentially of three parts.

(i) Prism table
(ii) collimator and
(iii) telescope.

Adjustment: Before using the spectrometer, the following adjustments must be made.

A. The axis of the telescope and that of the collimator must intersect the principal vertical axis of rotation of the telescope.

This adjustment is already being done by the manufacturer and can only be tested in the laboratory. For this purpose, a pin is mounted vertically in the center of prism table, and its image is observed in the telescope tube without eyepiece and for a wide slit in the collimator. If the image appears in the middle, at every position of telescope then the adjustment is perfect. Otherwise, the image is made in the center by using the screws supporting the telescope and collimator till the pins appear in the middle.

B. Prism table should be leveled

1. The prism table is leveled with help of three screws supporting the prism table. This is done with the help of spirit level.
2. The second method which is generally used is optical leveling of the prism table.

C. Telescope and collimator are adjusted for parallel light by Schuster's method.

First, prism is placed on the prism table and then adjusted approximately for minimum deviation position (see procedure). The spectrum is now seen through the telescope. Now the refracting edge of the prism is slightly turned towards the collimator, and the collimator is adjusted with the help of its rack and pinion arrangement such that spectrum is distinctly seen. Then the refracting edge of the prism is slightly turned towards the telescope, and telescope is adjusted with the help of its rack and pinion arrangement such that again the spectrum is distinctly seen. The process is repeated till the slight rotation of the prism table does not make the image go out of focus. This means that both collimator and telescope are now individually set for parallel rays.

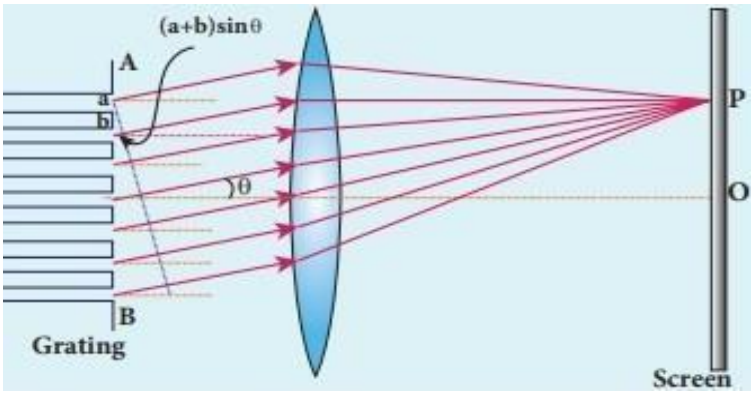


Figure 5.1(a): Diffraction of a plane wave incident normally on a multiple slit.

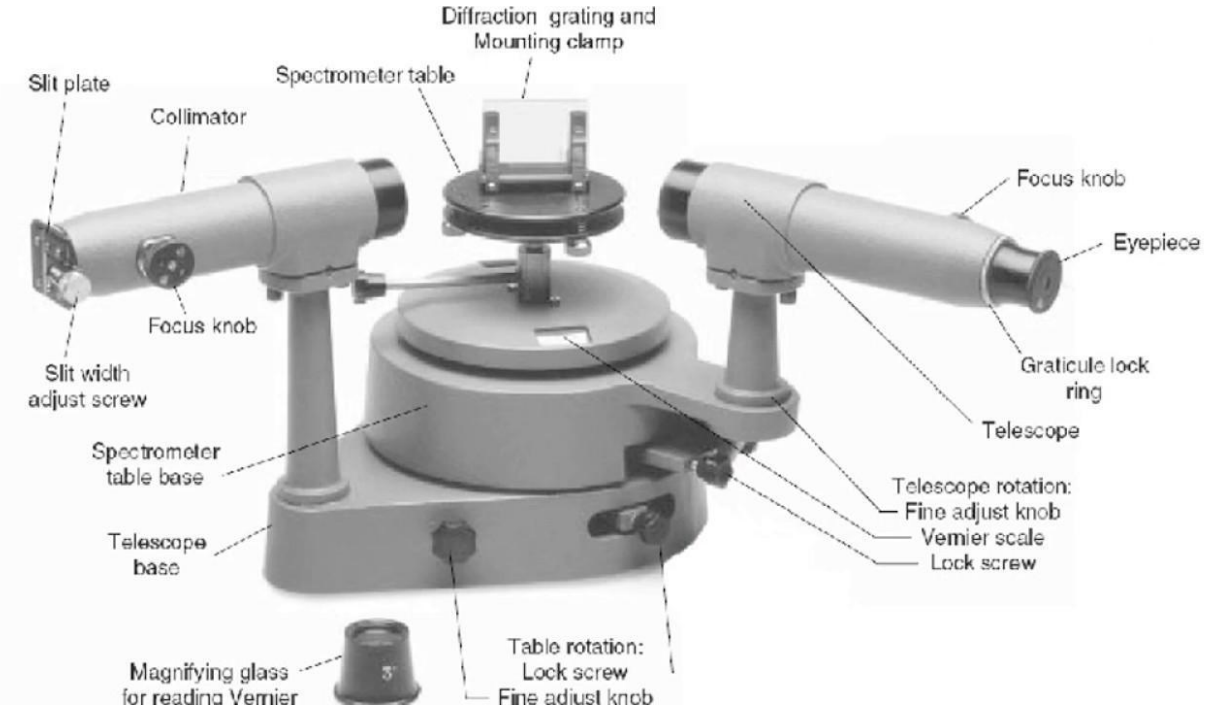


Figure 5.1 (b): Grating Spectrometer

5.4. Procedure:

1. Before the actual measurement of the angle of diffraction, the adjustment of spectrometer is made as given above.
2. Collimator and telescope are arranged in a line such that direct image of slit is visible. Coincide with the vertical cross wire on the direct image of slit and note the reading of the vernier scale.
3. Rotate the telescope by 90^0.
4. Rotate the grating table such that the slit image is visible in the telescope. coincide the vertical cross wire on the image and note the reading of vernier.
5. Turn the prism table by 45^0 or 135^0. In this situation grating is normal to the incident beam.

6. Slit should be adjusted parallel to the lines of the grating. For this setting, the slit is rotated in its own plane till the spectral lines become very sharp and bright.
7. Rotate the telescope to the left side of direct beam image and see the 1st order spectrum. Adjust and superimpose the vertical cross wire on different spectral lines (i.e. red, yellow and violet). Note the reading of Vernier for different spectral lines.
8. Find the diffraction angle for colours.

5.5. Observations

Observation table:

For the determination of the angle of diffraction

Number of lines on the grating (N) = 15000 lines/inch.

Value of the one division on the main scale of spectrometer = …….. Total divisions on the Vernier scale of spectrometer =……

$$Least\ count\ = \frac{Value\ of\ the\ 1\ division\ on\ the\ main\ scale}{Total\ number\ of\ divisions\ on\ the\ vernier\ scale} = _____$$

Order of the spectrum (n)	Colour of the line	Spectrum on the left of direct image (in degree)			Spectrum on the right of direct image (in degree)			Difference of the two readings of the same (2θ) (a-b)	Avg. Value of (2θ)	Diffraction angle for colour (θ)
		M S	V S	**Total (a)**	M S	V S	**Total (b)**			
n = 1	Violet V1									θ_V
	V2									
	Red V1									θ_R
	V2									

Standard wavelength values of colors

Violet - 4046 Å, Red - 6800 Å

5.6. Calculation:

$$esin\theta = n\lambda$$

here, *e* is the grating element = $\frac{2.54}{N}$, where N is the number of lines on the grating (N).

5.7. Results:

- The wavelength of violet line =..... Å
- The wavelength of red line =..... Å

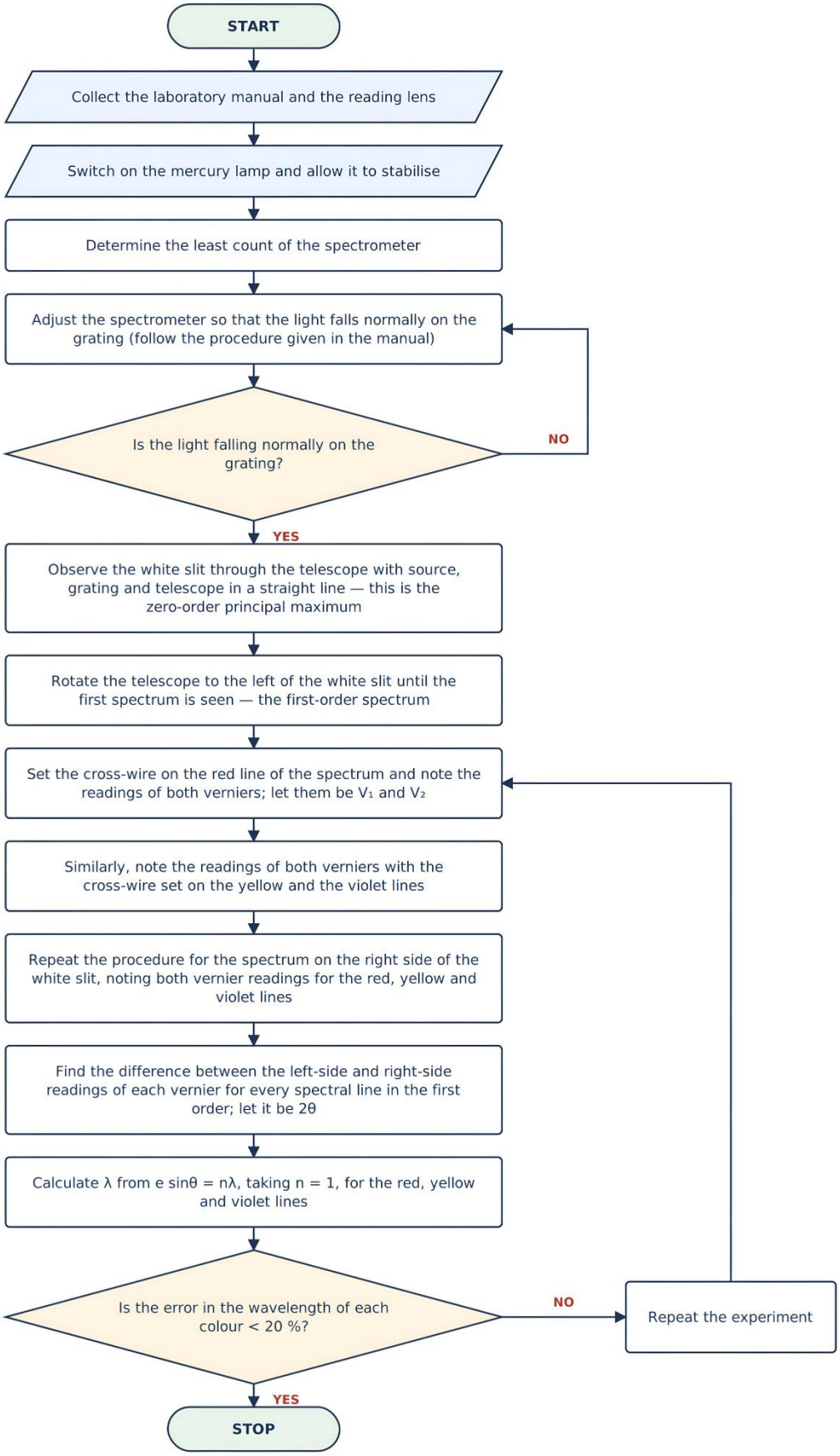
Percentage error:

$$\% \text{ error} = \frac{Std\ Value - Expt\ Value}{StdValue} \times 100 =$$

5.8. Precautionary measures to be taken:

1. Both the telescope and the collimator should be focused for parallel rays.
2. The eyepiece should be focused on the cross-wires.
3. The Prism table should be properly leveled.
4. While taking observations, slit should be made narrow and the cross-wires as a multiplication mark should be adjusted at the center of the image.

Exp. 5. Flow Chart



5.9. Applications of Spectrometer Experiment

1. Calibration of spectrometers.
2. Identification of unknown elements.
3. Study of atomic spectra.
4. Determination of energy levels.
5. Wavelength calibration of optical instruments.

5.10. Some sample viva-voice questions:

1. What is the principle of experiment and its condition to see pattern
2. by bare eyes?
3. What is Grating element?
4. What is Diffraction Grating?
5. Define essential parts of spectrometer.
6. What is the main function of diffraction grating?
7. Instead of mercury source if we use sodium vapor lamp, what changes can we observe?
8. What do you mean by diffraction?
9. 1 inch = …….. cm.
10. What is the lease count of a spectrometer?
11. Why does red deviate the most in case of grating?

Virtual Lab web link

1. Introduction: For calculation of Least Count.
   a. Vernier Calliper: https://www.olabs.edu.in/?sub=1&brch=5&sim=16&cnt=4
   b. Screw Gauge: https://www.olabs.edu.in/?sub=1&brch=5&sim=156&cnt=4
   c. Spectrometer: https://ov-amrt.vlabs.ac.in/exp/diffraction-grating/theory.html

2. Cyclotron: https://virtuelle-experimente.de/en/b-feld/anwendung/zyklotron.php

3. Numerical Aperture: https://lo-amrt.vlabs.ac.in/exp/numerical-aperture-optical-fiber/index.html
   (https://hmw-amrt.vlabs.ac.in/exp/meldes-string-apparatus/simulation/flash/flashplayer.exe)
   (https://hmw-amrt.vlabs.ac.in/exp/meldes-string-apparatus/simulation/flash/simulation/index.swf)

4. Newton's Ring: https://lo-amrt.vlabs.ac.in/exp/newtons-rings-wavelength-light/index.html

5. Wavelength using grating: https://ov-amrt.vlabs.ac.in/exp/diffraction-grating/theory.html

6. Planck's Constant: https://mp-amrt.vlabs.ac.in/exp/determination-plancks-constant/index.html

7. Melde's Exp.: https://hmw-amrt.vlabs.ac.in/exp/meldes-string-apparatus/index.html

8. Hall Effect: https://ssp-amrt.vlabs.ac.in/exp/hall-effect-charge-carrier/index.html